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Scientists find signs of extreme aging and youth in the same 117-year-old

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Source: Josep Carreras Leukaemia Research Institute

Summary: A remarkably detailed study of Maria Branyas, who lived past 117, reveals that extreme longevity may involve both intense aging and powerful biological protection at the same time. Researchers found signs of advanced aging, but also low inflammation, beneficial gut bacteria, protective genetic features, and a biological age younger than her chronological age. Because she avoided major disease, the study offered a rare look at aging itself rather than the illnesses that often accompany it.

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FULL STORY



Maria Branyas, the longest-lived person ever recorded, together with Dr. Manel Esteller, from the Josep Carreras Leukaemia Research Institute. Credit: Josep Carreras Leukaemia Research Institute

For roughly two centuries, human life expectancy climbed steadily as medicine, sanitation, nutrition, and healthcare improved. More recent research, however, suggests that this long upward trend may now be leveling off in developed countries. If that is the case, the next major challenge may no longer be simply preventing disease, but understanding aging itself.

That raises a fundamental question: what actually happens to the body as it ages, and could those biological changes someday be treated in much the same way as disease?

A Rare Look at Extreme Human Longevity

A research team led by Dr. Manel Esteller, Head of the Cancer Epigenetics group at the Josep Carreras Leukaemia Research Institute, has now published the final peer-reviewed results from an unusually detailed study of Maria Branyas, the Catalan woman who lived beyond 117 years.

Using samples collected with minimally invasive techniques, the researchers examined her biology across several layers at once. Their multi-omic analysis included genomic, proteomic, epigenomic, metabolomic and microbiomic technologies, making it the most comprehensive biological investigation yet conducted on a supercentenarian.

The findings, published in *Cell Reports Medicine*, came from an international and multidisciplinary team coordinated by Dr. Esteller and led by Eloy Santos.

Rather than finding that Branyas had simply aged more slowly across the board, the researchers uncovered a much more complicated picture.

As Esteller described it, her biology showed a "fascinating duality: the simultaneous presence of signals of extreme aging and of healthy longevity."

Signs of Aging Alongside Unusual Protection

Some parts of Branyas's biology clearly reflected her extraordinary age. The researchers found very short telomeres (the ends of chromosomes), an immune system with pro-inflammatory characteristics, and an aged population of B lymphocytes.

At the same time, other features appeared unusually favorable.

She carried genetic characteristics associated with neuroprotection and cardioprotection. She also had genuinely low inflammatory levels, a gut microbiome dominated by beneficial bifidobacteria, and an epigenetic biological age that was younger than her chronological age.

That combination is important because it suggests that reaching an exceptionally old age may not require escaping aging altogether. Instead, some people may experience pronounced signs of aging while also retaining biological traits that help protect against many of its harmful consequences.

Clues for Blood Cancers and Age-Related Disease



Aging in the blood-forming system is closely associated with a greater risk of serious and often incurable blood cancers, including leukemia and myelodysplastic syndromes.

Because of that connection, the biological insights from Branyas may have implications beyond longevity research. They could also help scientists better understand how these hematological conditions develop and why some older people are more vulnerable than others.

Separating Aging From Disease

One of the most unusual aspects of the study was Branyas's lack of serious disease.

That allowed researchers to examine the biological effects of aging with less interference from major illnesses, making this the first study in which aging could be clearly distinguished from disease in a person of such extreme age.

The findings therefore offer a rare, broad view of what aging itself can do to the human body and may eventually help identify ways to counter some of those changes.

Researchers caution that it is still too early to connect specific biological traits directly to particular behaviors or lifestyle choices. Even so, they point to several factors that may be worth considering when trying to understand Mrs. Branyas's exceptional longevity, including a healthy diet, a stimulating and diverse social network, and the absence of toxic habits.

Could Aging Itself Become Treatable?

By providing such a detailed biological snapshot of extreme longevity, the study gives researchers around the world a new reference point for investigating the aging process.

Ultimately, scientists hope that a clearer understanding of aging could lead to strategies that target it directly, much as modern medicine targets specific diseases.

Some of the tools needed for that effort may already exist. Epigenetic therapies and drugs designed specifically to combat senescence are already being explored in oncology, and both areas are closely connected to biological aging.

It remains uncertain whether these approaches could eventually be used to extend healthy human life. But if life expectancy has truly reached a plateau, therapies aimed at the biology of aging itself could become one of the paths scientists explore in an effort to push that limit higher.

This research has received public funding from the Generalitat de Catalunya, the European Community, and the Spanish Ministry of Science, Innovation and Universities, as well as private support from the "la Caixa" Foundation, the Cellex Foundation, the Spanish Association Against Cancer, and the John and Lucille Van Geest Foundation.

In addition to being accredited as a Severo Ochoa Center of Excellence by the Spanish Ministry of Science, Innovation and Universities – State Research Agency, the Josep Carreras Institute is also a CERCA, a research center of excellence of the Generalitat de Catalunya, and is accredited by the Scientific Foundation of the Spanish Association Against Cancer.

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Materials provided by **Josep Carreras Leukaemia Research Institute**. *Note: Content may be edited for style and length.*

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